

```
#include <stdio.h>
int main()
{
    int arr[100],n,i,max,min;
    printf("enter the number of elements");
    scanf("%d",&n)
    printf("enter the elements");
    for (i=0;i<n;i++)
    {
        scanf("%d",&arr[i]);
    }
    max=arr[0];
    min=arr[0];
    for (i=1;i<n;i++)
    {
        if (arr[i]>max){
            max=arr[i];
        }
        if (arr[i]<min){
            min=arr[i];
        }
    }
    printf("maximum=%d\n",max);
    printf("minimum=%d\n",min);
    return 0;
}
```

^G Get Help
^X Exit

^O WriteOut
^J Justify

^R Read File
^W Where is

^Y Prev Pg
^V Next Pg

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^L UnCut Text

^C Cur Pos
^T To Spell



```
Last login: Tue Sep 15 12:27:48 on ttys000
aditi@aditis-MacBook-Neo ~ % gcc maxmin.c -opk
aditi@aditis-MacBook-Neo ~ % ./pk
enter the number of elements 5
enter the elements 12 3 5 67 89
maximum=89
minimum=3
aditi@aditis-MacBook-Neo ~ %
```



```
#include <stdio.h>
int main()
{
    int x[100],n,i,even=0,odd=0;
    printf("enter the number of elements");
    scanf("%d",&n);
    printf("enter the elements");
    for (i=0;i<n;i++)
    {
        scanf("%d",&x[i]);
        if (x[i]%2==0){
            even++
        }else{
            odd++
        }
    }
    printf("odd numbers=%d\n",odd);
    printf("even numbers=%d\n",even);
    return 0;
}
```

^G Get Help
^X Exit

^O WriteOut
^J Justify

^R Read File
^W Where is

^Y Prev Pg
^V Next Pg

^K Cut Text
^U UnCut Text

^C Cur Pos
^T To Spell



```
Last login: Tue Sep 15 12:49:09 on ttys000
aditi@aditis-MacBook-Neo ~ % gcc evenodd.c
aditi@aditis-MacBook-Neo ~ % gcc evenodd.c -opl
aditi@aditis-MacBook-Neo ~ % ./pl
enter the number of elements 6
enter the elements 1 2 3 4 5 6
odd numbers=3
even numbers=3
aditi@aditis-MacBook-Neo ~ % █
```



```
#include <stdio.h>
int main()
{
    int ad[100],n,i,positive=0,negative=0,zero=0;
    printf("enter the no of elements");
    scanf("%d",&n);
    printf("enter the elements");
    for (i=0;i<n;i++)
    {
        scanf("%d",&ad[i]);
        if (ad[i]>0){
            positive++;
        }else if (ad[i]<0){
            negative++;
        }else{
            zero++;
        }
    }
    printf("positive numbers=%d\n",positive);
    printf("negative numbers=%d\n",negative);
    printf("zero=%d\n",zero);
    return 0;
}
```

^G Get Help
^X Exit

^O WriteOut
^J Justify

^R Read File
^W Where is

^Y Prev Pg
^V Next Pg

^K Cut Text
^L UnCut Text

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^T To Spell



```
Last login: Tue Sep 15 12:52:01 on ttys000
aditi@aditis-MacBook-Neo ~ % gcc positivenegative.c -o pl
aditi@aditis-MacBook-Neo ~ % ./pl
enter the no of elements6
enter the elements 10 -5 0 8 -2 0
positive numbers=2
negative numbers=2
zero=2
aditi@aditis-MacBook-Neo ~ %
```



```
#include <stdio.h>
int main()
{
    int a[100],n,i,key,found=0;
    printf("enter the no of elements");
    scanf("%d",&n);
    printf("enter the elements");
    for (i=0;i<n;i++)
    {
        scanf("%d",&a[i]);
    }
    printf("enter element to search");
    scanf("%d",&key);
    for (i=0;i<n;i++)
    {
        if (a[i]==key){
            found=1;
            printf("element found at position %d",i);
            break;
        }
    }
    if (found==0){
        printf("element not found");
    }
    return 0;
}
```

 Get Help
 Exit

 WriteOut
 Justify

 Read File
 Where is

 Prev Pg
 Next Pg

 Cut Text
 UnCut Text

 Cur Pos
 To Spell



```
Last login: Tue Sep 15 12:52:01 on ttys000
aditi@aditis-MacBook-Neo ~ % gcc positivenegative.c -o pl
aditi@aditis-MacBook-Neo ~ % ./pl
enter the no of elements 6
enter the elements 10 -5 0 8 -2 0
positive numbers=2
negative numbers=2
zero=2
aditi@aditis-MacBook-Neo ~ % nano linearsearch.c
aditi@aditis-MacBook-Neo ~ % gcc linearsearch.c -o pl
aditi@aditis-MacBook-Neo ~ % ./pl
enter the no of elements 6
enter the elements 10 20 30 40 50 60
enter element to search 30
element found at position 2
aditi@aditis-MacBook-Neo ~ %
```



```
#include <stdio.h>
int main()
{
    int a[100],n,i,temp;
    printf("enter the number of elements");
    scanf("%d",&n);
    printf("enter the elements");
    for (i=0;i<n;i++)
    {
        scanf("%d",&a[i]);
    }
    for (i=0;i<n/2;i++)
    {
        temp=a[i];
        a[i]=a[n-1-i];
        a[n-1-i]=temp;
    }
    printf("reversed array");
    for (i=0;i<n;i++){
        printf("%d",a[i]);
    }
    return 0;
}
```

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^X Exit

^O WriteOut
^J Justify

^R Read File
^W Where is

^Y Prev Pg
^V Next Pg

^K Cut Text
^U UnCut Text

^C Cur Pos
^T To Spell



```
Last login: Tue Sep 15 12:52:01 on ttys000
aditi@aditis-MacBook-Neo ~ % gcc positivenegative.c -opl
aditi@aditis-MacBook-Neo ~ % ./pl
enter the no of elements6
enter the elements 10 -5 0 8 -2 0
positive numbers=2
negative numbers=2
zero=2
aditi@aditis-MacBook-Neo ~ % nano linearsearch.c
aditi@aditis-MacBook-Neo ~ % gcc linearsearch.c -opl
aditi@aditis-MacBook-Neo ~ % ./pl
enter the no of elements 6
enter the elements10 20 30 40 50 60
enter element to search 30
element found at position 2
aditi@aditis-MacBook-Neo ~ % nano reversearray.c\
>
aditi@aditis-MacBook-Neo ~ % gcc reversearray.c -opl
aditi@aditis-MacBook-Neo ~ % ./pl
enter the number of elements5
enter the elements10 89 78 67 44
reversed array4467788910
aditi@aditis-MacBook-Neo ~ %
```



```
#include <stdio.h>
int main()
{
    int array[100],n,i,sum=0;
    printf("enter the number of elements");
    scanf("%d",&n);
    printf("enter the elements");
    for (i=0;i<n;i++)
    {
        scanf("%d",&array[i]);
        sum=sum+array[i];
    }
    printf("sum=%d\n",sum);
    return 0;
}
```

^G Get Help
^X Exit

^O WriteOut
^J Justify

^R Read File
^W Where is

^Y Prev Pg
^V Next Pg

^K Cut Text
^U UnCut Text

^C Cur Pos
^T To Spell



```
Last login: Tue Sep 15 12:27:07 on ttys000
aditi@aditis-MacBook-Neo ~ % gcc sumofarray.c -opk
aditi@aditis-MacBook-Neo ~ % ./pk
enter the number of elements 5
enter the elements 10 20 30 40 50
sum=150
aditi@aditis-MacBook-Neo ~ % █
```



```
#include <stdio.h>
int main()
{
    int n,digit,i;
    int count[10]={0};
    int max=0,result=0;
    printf("enter a integer");
    scanf("%d",&n);
    if (n==0){
        count[0]=1;
    }
    if (n<0)
    {
        n=-n;
    }
    while (n>0){
        digit=n%10;
        count[digit]++;
        n=n/10;
    }
    for (i=0;i<10;i++){
        if (count[i]>max){
            max=count[i];
            result=i;
        }
    }
    printf("most frequent digit=%d\n",result);
    printf("occurence=%d",max);
    return 0;
}
```

 Get Help
 Exit

 WriteOut
 Justify

 Read File
 Where is

 Prev Pg
 Next Pg

 Cut Text
 UnCut Text

 Cur Pos
 To Spell



```
Last login: Tue Sep 15 13:01:49 on ttys000
aditi@aditis-MacBook-Neo ~ % gcc merge.c -o pl
aditi@aditis-MacBook-Neo ~ % ./pl
enter the size of first array 3
enter first array elements 1 4 5
enter the size of second array 3
enter second array elements 3 4 5
merged array:145345
aditi@aditis-MacBook-Neo ~ % nano mosttimesinarray.c
aditi@aditis-MacBook-Neo ~ % gcc mosttimesinarray.c -o pl
aditi@aditis-MacBook-Neo ~ % ./pl
enter a integer 122334
most frequent digit=2
occurrence=2
aditi@aditis-MacBook-Neo ~ %
```



```
#include <stdio.h>
int main()
{
    int a[100],n,i,key;
    int low,high,mid;
    int found=0;
    printf("enter number of elements");
    scanf("%d",&n);
    printf("enter sorted elements");
    for (i=0;i<n;i++)
    {
        scanf("%d",&a[i]);
    }
    printf("enter element to search:");
    scanf("%d",&key);
    low=0;
    high=n-1;
    while (low<=high){
        mid=(low+high)/2;
        if (a[mid]==key){
            found=1;
            printf("elements found at position %d",mid);
            break;
        }else if (key>a[mid]){
            low=mid+1;
        }else{
            high=mid-1;
        }
    }
    if (found==0){
        printf("elements not found");
    }
    return 0;
}
```

^G Get Help
^X Exit

^O WriteOut
^J Justify

^R Read File
^K Where is

^V Prev Pg
^V Next Pg

^K Cut Text
^U UnCut Text

^C Cur Pos
^T To Spell



```
Last login: Tue Sep 15 13:26:31 on ttys000
aditi@aditis-MacBook-Neo ~ % nano binarysearch.c
aditi@aditis-MacBook-Neo ~ % gcc binarysearch.c -opk
aditi@aditis-MacBook-Neo ~ % ./pk
enter number of elements 5
enter sorted elements10 20 30 40 50
enter element to search: 40
elements found at position 3
aditi@aditis-MacBook-Neo ~ %
```



```
#include <stdio.h>
int main()
{
    int a[100],n,i,value,pos;
    printf("enter number of elements");
    scanf("%d",&n);
    printf("enter sorted elements");
    for (i=0;i<n;i++){
        scanf("%d",&a[i]);
    }
    printf("enter elements to insert");
    scanf("%d",&value);
    pos=n;
    for(i=0;i<n;i++){
        if (value<a[i]){
            pos=i;
            break;
        }
    }
    for (i=n;i>pos;i--){
        a[i]=a[i-1];
    }
    a[pos]=value;
    n++;
    printf("array after insertion");
    for (i=0;i<n;i++){
        printf("%d",a[i]);
    }
    return 0;
}
```

^G Get Help
^X Exit

^O WriteOut
^J Justify

^R Read File
^W Where is

^V Prev Pg
^V Next Pg

^K Cut Text
^U UnCut Text

^C Cur Pos
^T To Spell



```
Last login: Tue Sep 15 18:50:15 on ttys000
aditi@aditis-MacBook-Neo ~ % gcc insertelement.c -o pl
aditi@aditis-MacBook-Neo ~ % ./pl
enter number of elements 5
enter sorted elements 10 20 30 40 50
enter elements to insert 67
array after insertion 10 20 30 40 50 67
aditi@aditis-MacBook-Neo ~ %
```



```
#include <stdio.h>
int main()
{
    int array[100],n,i;
    printf("enter the number of elements");
    scanf("%d",&n);
    printf("enter the elements\n");
    for (i=0;i<n;i++)
    {
        scanf("%d",&array[i]);
    }
    printf("array elements are");
    for (i=0;i<n;i++)
    {
        printf("%d",array[i]);
    }
    return 0;
}
```

^G Get Help
^X Exit

^O WriteOut
^D Justify

^R Read File
^W Where is

^V Prev Pg
^V Next Pg

^K Cut Text
^U UnCut Text

^C Cur Pos
^T To Spell



```
Last login: Sat Sep 12 15:03:36 on ttys000
aditi@aditis-MacBook-Neo ~ % nano print1d.c
aditi@aditis-MacBook-Neo ~ % gcc print1d.c -opk
aditi@aditis-MacBook-Neo ~ % ./pk
enter the number of elements 5
enter the elements
10 20 30 40 50
array elements are1020304050%
aditi@aditis-MacBook-Neo ~ %
```



```
#include <stdio.h>
int main()
{
    int a[100],n,i,value,pos;
    printf("enter the number of elements");
    scanf("%d",&n);
    printf("enter elements\n");
    for (i=0;i<n;i++){
        scanf("%d",&a[i]);
    }
    printf("enter the position");
    scanf("%d",&pos);
    printf("enter the element to insert");
    scanf("%d",&value);
    for (i=n;i>pos;i--){
        a[i]=a[i-1];
    }
    a[pos]=value;
    n++;
    printf("array after insertion");
    for (i=0;i<n;i++){
        printf("%d",a[i]);
    }
    return 0;
}
```

^G Get Help
^X Exit

^O WriteOut
^J Justify

^R Read File
^W where is

^Y Prev Pg
^V Next Pg

^K Cut Text
^U UnCut Text

^C Cur Pos
^T To Spell



```
#include <stdio.h>
int main()
{
    int a[100],n,i,pos;
    printf("enter the number of elements");
    scanf("%d",&n);
    printf("enter elements\n");
    for (i=0;i<n;i++){
        scanf("%d",&a[i]);
    }
    printf("enter position to delete");
    scanf("%d",&pos);
    for (i=pos;i<n-1;i++){
        a[i]=a[i+1];
    }
    n--;
    printf("array after deletion");
    for (i=0;i<n;i++){
        printf("%d",a[i]);
    }
    return 0;
}
```

^G Get Help
^X Exit

^O WriteOut
^J Justify

^R Read File
^W Where is

^V Prev Pg
^V Next Pg

^K Cut Text
^U UnCut Text

^C Cur Pos
^T To Spell



```
Last login: Tue Sep 15 19:17:25 on ttys000
aditi@aditi-MacBook-Neo ~ % gcc deleteelement.c -o pd
aditi@aditi-MacBook-Neo ~ % ./pd
enter the number of elements 5
enter elements
10 30 56 780 89
enter position to delete 2
array after deletion103078089
aditi@aditi-MacBook-Neo ~ %
```



```
#include <stdio.h>
#include <limits.h>

int main() {
    int a[100], n, i;
    int largest, second;
    int found = 0;
    printf("Enter number of elements: ");
    scanf("%d", &n);
    printf("Enter elements:\n");
    for (i = 0; i < n; i++) {
        scanf("%d", &a[i]);
    }
    largest = a[0];
    second = INT_MIN;
    for (i = 1; i < n; i++) {
        if (a[i] > largest) {
            second = largest;
            largest = a[i];
        } else if (a[i] > second && a[i] != largest) {
            second = a[i];
        }
    }
    for (i = 0; i < n; i++) {
        if (a[i] != largest) {
            found = 1;
            break;
        }
    }
    if (found == 1) {
        printf("Second largest element = %d", second);
    } else {
        printf("Second largest element does not exist");
    }
    return 0;
}
```

^G Get Help
^X Exit

^O WriteOut
^J Justify

^R Read File
^W Where is

^Y Prev Pg
^V Next Pg

^K Cut Text
^U UnCut Text

^C Cur Pos
^T To Spell



```
Last login: Tue Sep 15 19:25:33 on ttys000
aditi@aditis-MacBook-Neo ~ % gcc secondlargest.c -opk
aditi@aditis-MacBook-Neo ~ % ./pk
Enter number of elements: 5
Enter elements:
10 30 78 56 98
Second largest element = 78
aditi@aditis-MacBook-Neo ~ %
```



```
#include <stdio.h>
int main()
{
    int a[100],b[100],c[200];
    int n1,n2,i;
    printf("enter the size of first array");
    scanf("%d",&n1);
    printf("enter first array elements");
    for (i=0;i<n1;i++)
    {
        scanf("%d",&a[i]);
        c[i]=a[i];
    }
    printf("enter the size of second array");
    scanf("%d",&n2);
    printf("enter second array elements");
    for (i=0;i<n2;i++)
    {
        scanf("%d",&b[i]);
        c[n1+i]=b[i];
    }
    printf("merged array:");
    for (i=0;i<n1+n2;i++)
    {
        printf("%d",c[i]);
    }
    return 0;
}
```

^G Get Help
^X Exit

^O WriteOut
^J Justify

^R Read File
^W Where is

^Y Prev Pg
^V Next Pg

^K Cut Text
^U UnCut Text

^C Cur Pos
^T To Spell



```
Last login: Tue Sep 15 19:29:55 on ttys000
aditi@aditis-MacBook-Neo ~ % nano merge.c
aditi@aditis-MacBook-Neo ~ % gcc merge.c -opk
aditi@aditis-MacBook-Neo ~ % ./pk
enter the size of first array 3
enter first array elements 10 20 30
enter the size of second array 3
enter second array elements 40 50 60
merged array:102030405060
aditi@aditis-MacBook-Neo ~ %
```



```
Last login: Tue Sep 15 19:29:55 on ttys000
aditi@aditis-MacBook-Neo ~ % nano merge.c
aditi@aditis-MacBook-Neo ~ % gcc merge.c -opk
aditi@aditis-MacBook-Neo ~ % ./pk
enter the size of first array 3
enter first array elements 10 20 30
enter the size of second array 3
enter second array elements 40 50 60
merged array:102030405060
aditi@aditis-MacBook-Neo ~ %
```

